07-31 第1篇

阅读理解（说明文）

How Sound Travels

① When you clap your hands or someone calls your name, you hear a sound. But have you ever thought about how sound reaches your ears? Sound is a kind of energy that moves through the air, water, or other materials. Understanding how sound travels helps us explain many things in our daily life.

② Sound is made when something vibrates (振动). For example, when you hit a drum, the drum's surface moves quickly back and forth. This movement pushes the air around it. The air molecules (分子) then bump into each other and pass the movement along, much like a wave in the water. These waves travel outward in all directions until they reach your ears.

③ Sound travels at different speeds through different materials. In the air at normal temperature, sound travels at about 343 meters per second. It moves faster through water — about 1,500 meters per second. Sound travels even faster through solid materials like wood or metal. That is why when you put your ear against a long table and tap it at one end, you can hear the tap almost at once at the other end.

④ Sound cannot travel through empty space. There is no air or other material in space for the sound waves to move through. So if you were floating in space, you would not hear anything at all — not even a huge explosion. This is why astronauts use radios to talk to each other.

⑤ We use the way sound travels in many ways. Doctors use it to check our bodies. Bats use it to find their way in the dark. Ships use it to find objects under the sea. The more we learn about sound, the more we can use it to make our lives better.

注释： vibrate 振动；molecule 分子；wave 波；astronaut 宇航员

选择题

1. According to paragraph ②, how is sound produced? A. When something vibrates and the movement is passed through the air like a wave. B. When air moves very slowly. C. When we clap our hands only. D. When water waves hit the shore.

2. Where does sound travel fastest? A. Through the air. B. Through empty space. C. Through solid materials like wood or metal. D. Through water.

3. Why cannot we hear sound in space? A. Because space is too cold. B. Because the sun is too bright. C. Because astronauts do not make any sound. D. Because there is no air or material for sound waves to travel through.

4. What can we infer from paragraph ③ about the table experiment? A. The table makes the sound louder. B. Sound travels faster through the solid table than through the air. C. Tapping the table creates no sound. D. The experiment only works with wooden tables.

5. What is the main idea of this passage? A. Drums are the best way to make sound. B. Sound travels as waves through materials, and we use this knowledge in many ways. C. Space is a dangerous place for humans. D. Bats are the smartest animals on Earth.

详细解析

1. 【答案】A

【题型】 细节理解

【定位】 第②段 "Sound is made when something vibrates... This movement pushes the air around it. The air molecules then bump into each other and pass the movement along, much like a wave in the water."

【解析】 原文明确说明声音由物体振动产生，振动推动空气，空气分子相互碰撞并像水波一样传递这种运动。A项准确概括。B项"空气缓慢移动"与原文矛盾；C项"仅拍手"过于片面；D项"水波拍岸"属于偷换概念。故选A。

2. 【答案】C

【题型】 细节理解

【定位】 第③段 "Sound travels at different speeds through different materials... In the air... about 343 meters per second. It moves faster through water — about 1,500 meters per second. Sound travels even faster through solid materials like wood or metal."

【解析】 原文依次给出空气中约343米/秒、水中约1500米/秒、固体（如木头或金属）中更快，故固体中声速最快。C项正确。A、D项速度均低于固体；B项"太空"中声音无法传播。故选C。

3. 【答案】D

【题型】 细节理解

【定位】 第④段 "Sound cannot travel through empty space. There is no air or other material in space for the sound waves to move through."

【解析】 原文清楚说明声音无法在真空中传播，因为太空中没有空气或其他介质供声波传播。D项准确复述。A项"太冷"、B项"太阳太亮"、C项"宇航员不发声"均与原文因果不符。故选D。

4. 【答案】A

【题型】 推理判断

【定位】 第③段 "Sound travels even faster through solid materials like wood or metal. That is why when you put your ear against a long table and tap it at one end, you can hear the tap almost at once at the other end."

【解析】 原文通过桌上敲击实验说明固体传声更快，可推知声音通过固体桌子的传播速度大于通过空气。A项正确。B项"使声音更大"未提及；C项"不产生声音"与实验矛盾；D项"仅木桌有效"过于绝对。故选A。

5. 【答案】D

【题型】 主旨大意

【定位】 全文：第①段引入，第②段声的产生与传播，第③段不同介质中声速，第④段真空无法传声，第⑤段声的应用。

【解析】 全文围绕声音的传播原理展开，并说明该知识在医学、动物、航海等方面的应用，D项最能概括主旨。A项只涉及鼓；B项只涉及太空；C项只涉及蝙蝠，均以偏概全。故选D。